

KARTHIK DEVARAJ

Data & IoT Engineering Intern | Python, SQL, Embedded C | Data Analysis & Sensor Systems

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Professional Summary

Computer Applications student with hands-on experience in IoT systems, embedded programming, and web development. Proven ability to build functional prototypes including gas detection systems with GSM integration, RFID-based hospital management, and IoT plant monitoring solutions. Skilled in Python, Embedded C, Arduino, HTML/CSS/JavaScript, and SQL, with exposure to sensor data analysis. Seeking internship opportunities to apply technical skills in real-world engineering and data-driven environments.

Core Competencies & Expertise

- **Programming Languages:** C, Python, Embedded C, JavaScript, HTML5, CSS3, SQL
- **Embedded Systems & IoT:** Arduino IDE, Arduino Uno, Microcontrollers, Sensor Integration (MQ2 Smoke, Gas Detection, RFID), GSM Modules, UART, I2C Communication Protocols.
- **Web Technologies:** HTML, CSS, JavaScript, Responsive Web Design, Front-End Development, UI/UX Design.
- **Databases & Tools:** MySQL, SQL Queries, Git, VS Code, Data Structures & Algorithms (DSA)
- **Testing & Quality:** Manual Testing, Test Case Design, Bug Tracking, Root Cause Analysis, API Testing Concepts
- **Data Analysis:** Python for Data Analysis, Excel, SQL for Data Manipulation, Statistical Analysis Basics
- **Soft Skills:** Problem Solving, Analytical Thinking, Team Collaboration, Communication, Project Management.

Education

Bachelor of Computer Applications (BCA)

PES University, RR Campus, Bengaluru | Graduation: 2026

- **Academic Excellence:** Distinction Awardee for Semester 1, 4, and 5
- **Relevant Coursework:** Data Structures, Algorithms, IoT Systems, Database Management, Web Technologies, Software Engineering

Certifications & Achievements

- Academic Distinction Awardee | PES University | Semester 1, 4, and 5 (Top 10% of cohort)
- PESU I/O – Automation in IoT | PES University | Project-based certification program
- Front-End Web Development | Udemy | Comprehensive course covering HTML, CSS, JavaScript, and responsive design
- CIE Ignite Innovation Program – Stage 2 | Shortlisted among 200+ project submissions for innovation excellence

Technical Projects

1) Garden Genie – IoT-Based Plant Monitoring System

Technologies: *Arduino, Soil Moisture Sensors, Temperature Sensors, IoT Protocols, Mobile App Integration / Ongoing*

- Developing intelligent urban gardening solutions with real-time sensor-based plant health monitoring and automated alerts
- Implementing IoT device connectivity for mobile application enabling remote monitoring and data visualization
- Integrating AI chatbot functionality to provide automated plant care recommendations based on sensor data

2) Gas Leakage Detection & Emergency Alert System

Technologies: *Arduino, MQ-Series Gas Sensors, GSM Module (UART), LCD Display (I2C), Embedded C*

- Designed and built automated gas detection system with threshold-based alarm triggering for LPG and toxic gas detection
- Integrated GSM module using UART protocol for automated IVR call notifications to users during gas leak detection
- Implemented I2C communication protocol for real-time gas concentration display on 16x2 LCD panel
- Improved detection reliability through sensor calibration and threshold optimization

3) RFID-Based Hospital Management System

Technologies: *RFID (RC522), Arduino, Database Management (SQL), Embedded Systems / CodeBerries Hackathon*

- Developed RFID-based patient identification and tracking system for automated medical record retrieval
- Implemented database integration for real-time patient data storage and retrieval upon RFID tag scan
- Streamlined patient identification workflow and reduced manual lookup effort

4) Mimetic – AI-Powered Content Repurposing Platform

Technologies: *HTML5, CSS3, JavaScript, Responsive Design, UI/UX Wireframing*

- Collaborated on AI-driven content transformation tool enabling digital creators to repurpose content across platforms
- Designed and developed responsive front-end prototype with user-centric interface and seamless navigation
- Project shortlisted for Stage 2 of PES University CIE Ignite Innovation Program among 200+ submissions

5) Automated Smoke Detection & Alert System

Technologies: *Arduino Uno, MQ2 Smoke Sensor, Embedded C, Buzzer Alert System*

- Built smoke detection system using MQ2 sensor with Arduino microcontroller for early fire hazard warning
- Programmed threshold-based audible alert mechanism triggered upon detecting smoke concentration above safe limits
- Demonstrated understanding of sensor interfacing, analog-to-digital conversion, and safety system design principles

Professional Experience

Freelance Event Coordinator

Paytm Insider (District by Zomato) | Bengaluru | October 2024 - November 2025

- Coordinated 35+ large-scale live events with 10000+ attendees, ensuring seamless operations and adherence to event protocols. Performed data verification and quality checks on sales reports, identifying and resolving data mismatches to ensure integrity
- Managed ticketing systems and box-office operations achieving 99%+ transaction accuracy in high-pressure environments

Additional Information

- **Languages:** English (Fluent), Hindi (Professional), Kannada (Native)
- **Availability:** Immediate (Can join internship within 1 week)
- **Location Preference:** Bengaluru (Primary), Open to Remote/Hybrid opportunities
- **Interests:** IoT Innovation, Embedded Systems Design, Web Technologies, Data-Driven Solutions, Automation

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.

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